

COMP9334: Capacity Planning of Computer Systems and Networks

Optimisation (5):
Power of binary variables

Integer Programming - What have you seen?

- A recurrent theme is to use integer programming to make *binary decisions*
- Examples of binary decisions
 - Week 9A: Grid computing problem
 - Choose a particular grid computing company or not
 - Week 9B: Routing of flows
 - Should the flow be routed on a link or not?
 - Week 10A: Placement problem
 - Should a location be chosen or not?

This week's lecture: Power of binary variables

- Not only for making yes-or-no type of decisions, binary variables can be used to capture many other requirements
 - Restricted range of values
 - Either-or constraints
 - If-then constraints
 - Piecewise linear functions

Restricted range of values

- Some variables can only take certain values
 - E.g. network links can only be of capacity 155 Mbps, 466 Mbps, 622 Mbps, etc
- If decision variable x can only take values from $\{a_1, a_2, \dots, a_m\}$, this can be modeled by using an additional set of binary decision variables

$$y_i = \begin{cases} 1 & \text{if } a_i \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

Restricted range of values

- Then, the above requirement can be captured by

$$\begin{aligned}x &= \sum_{i=1}^m a_i y_i \\ \sum_{i=1}^m y_i &= 1 \\ y_i &\in \{0, 1\}\end{aligned}$$

- E.g. if $a_1 = 155, a_2 = 466, a_3 = 622$, we have

- $y_1 = 1 \Rightarrow y_2 = y_3 = 0 \Rightarrow x = 155$

- $y_2 = 1 \Rightarrow y_1 = y_3 = 0 \Rightarrow x = 466$

- $y_3 = 1 \Rightarrow y_1 = y_2 = 0 \Rightarrow x = 622$

Either-or constraints

- A Cloud computing service provider offers 3 different packages with different speed and cost for each package. You can buy any cycles from any package but the deal requires that
 - # cycles from Package 1 + # cycles from Package 2 ≥ 10000 , or,
 - # cycles from Package 2 + # cycles from Package 3 ≥ 50000
 - At least one of these two inequalities must hold, but not necessarily both
- Let w_i = number of cycles to be bought from Package i

Either-or constraints (cont.)

- The above requirement can be captured by using an additional binary decision variable p

$$\begin{aligned}w_1 + w_2 &\geq 10000p \\w_2 + w_3 &\geq 50000(1 - p) \\p &\in \{0, 1\} \\w_i &\geq 0, \quad i = 1, 2, 3\end{aligned}$$

Case 1: $p = 0$, we have

$$\begin{aligned}w_1 + w_2 &\geq 0 \leftarrow \text{Trivially satisfied} \\w_2 + w_3 &\geq 50000 \\w_i &\geq 0, \quad i = 1, 2, 3\end{aligned}$$

Case 2: $p = 1$, we have

$$\begin{aligned}w_1 + w_2 &\geq 10000 \\w_2 + w_3 &\geq 0 \leftarrow \text{Trivially satisfied} \\w_i &\geq 0, \quad i = 1, 2, 3\end{aligned}$$

Either-or constraints (cont.)

- In general, if one of the following two constraints must be satisfied

$$\begin{aligned}\sum_{i=1}^n a_{1,i}x_i &\geq b_1 \\ \sum_{i=1}^n a_{2,i}x_i &\geq b_2\end{aligned}$$

where $a_{j,i}$ are given parameters, $x_i (\geq 0)$ are decision variables, b_j 's are constants, then the either-or constraints can be modelled by

$$\begin{aligned}\sum_{i=1}^n a_{1,i}x_i &\geq b_1p \\ \sum_{i=1}^n a_{2,i}x_i &\geq b_2(1-p) \\ p &\in \{0, 1\}\end{aligned}$$

If-then constraints

- We may want to impose if-then constraints, e.g.

$$\text{if } x_1 + x_2 > 1, \text{ then } y \geq 4$$

where x_1, x_2 are binary variables, and $0 \leq y \leq 10$

- The above if-then constraint can be captured by using an additional binary decision variable p

$$\begin{aligned} x_1 + x_2 - 1 &\leq 1 - p \\ -y + 4 &\leq 4p \\ p &\in \{0, 1\} \end{aligned}$$

If-then constraints (cont.)

- To understand how this works, consider the two cases:
 - Case 1: If $x_1 + x_2 > 1$ holds
 - Since $x_1 + x_2 > 1$, $x_1 + x_2 - 1 > 0$
 - Since p can only be 1 or 0, the inequality constraint $x_1 + x_2 - 1 \leq 1 - p$ forces p to be 0
 - Since $p = 0$, from the inequality constraint $-y + 4 \leq 4p$, we have $y \geq 4$ which is the condition that we want to impose when $x_1 + x_2 > 1$ holds
 - Case 2: If $x_1 + x_2 > 1$ does not hold
 - In this case, since $x_1 + x_2 - 1 \leq 0$, p can be either 0 or 1
 - If $p = 0$, the inequality constraint $-y + 4 \leq 4p$ becomes $y \geq 4$
 - If $p = 1$, the inequality constraint $-y + 4 \leq 4p$ becomes $y \geq 0$
 - Thus, p can be chosen such that there is no restriction on the value of y

If-then constraints (cont.)

- In general, the if-then constraint

$$\text{if } f(x_1, x_2, \dots, x_n) > 0, \text{ then } g(x_1, x_2, \dots, x_n) \geq 0$$

can be modeled by

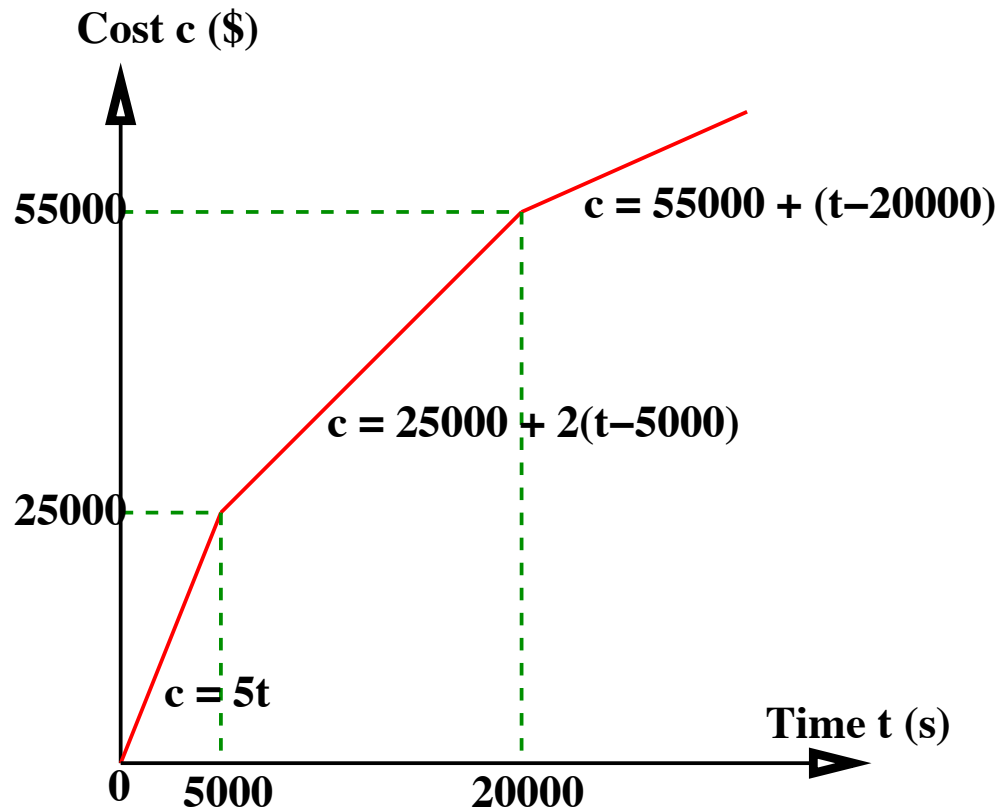
$$\begin{aligned} f(x_1, x_2, \dots, x_n) &\leq M_1(1 - p) \\ -g(x_1, x_2, \dots, x_n) &\leq M_2 p \end{aligned}$$

where p is a binary variable, M_1 and M_2 are constants chosen large enough such that $f(x_1, x_2, \dots, x_n) \leq M_1$ and $-g(x_1, x_2, \dots, x_n) \leq M_2$ hold for all possible choices of $x_1, x_2, \dots, x_n$

Piecewise linear functions

- We can use binary variables to model piecewise linear functions
- Example: A Cloud computing service provider may use a progressive charging scheme
 - 5 dollars/sec for the first 5,000 sec
 - 2 dollars/sec for the next 15,000 sec
 - 1 dollar/sec thereafter

Piecewise linear functions (cont.)



■ Decision variables

$$y_i = \begin{cases} 1 & \text{if segment } i \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

- Segment 1: $0 \leq t \leq 5000$, $\text{cost} = 5t$
- Segment 2: $5000 \leq t \leq 20000$, $\text{cost} = 2t + 15000$
- Segment 3: $20000 \leq t$, $\text{cost} = t + 35000$

Piecewise linear functions (cont.)

■ We have

- $y_1 = 1 \Rightarrow 0 \leq t \leq 5000$ and $\text{cost} = 5t$
- $y_2 = 1 \Rightarrow 5000 \leq t \leq 20000$ and $\text{cost} = 2t + 15000$
- $y_3 = 1 \Rightarrow 20000 \leq t$ and $\text{cost} = t + 35000$
- $y_1 + y_2 + y_3 = 1$

■ We can rewrite these as

- $0 \leq ty_1 \leq 5000y_1$
- $5000y_2 \leq ty_2 \leq 20000y_2$
- $20000y_3 \leq ty_3$
- $\text{cost} = y_1(5t) + y_2(2t + 15000) + y_3(t + 35000)$
- $y_1 + y_2 + y_3 = 1$

■ Problem: non-linear constraints

Piecewise linear functions (cont.)

- Define $t_i = ty_i$ for $i = 1, 2, 3$

- $\text{cost} = 5t_1 + 2t_2 + 15000y_2 + t_3 + 35000y_3$

- $0 \leq t_1 \leq 5000y_1$

- $5000y_2 \leq t_2 \leq 20000y_2$

- $20000y_3 \leq t_3 \leq My_3$

- $y_1 + y_2 + y_3 = 1$

- $t = t_1 + t_2 + t_3$

- Note

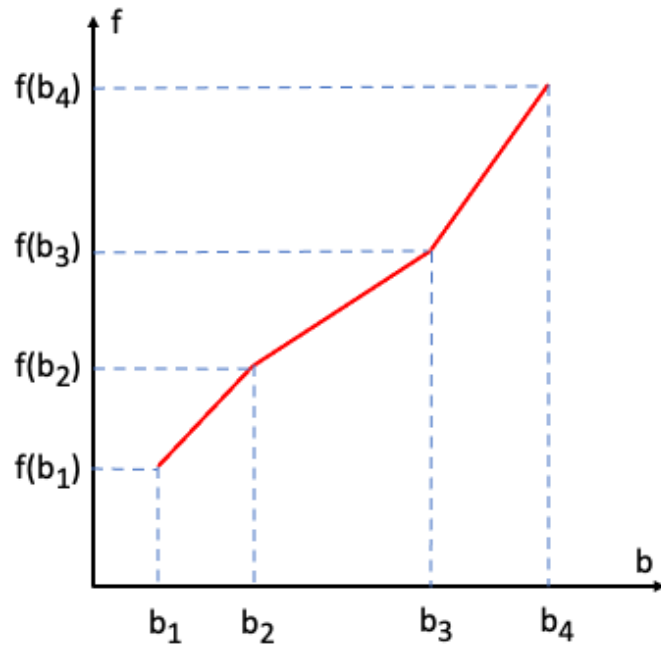
- t_i is non-zero if the corresponding $y_i = 1$

- M is a sufficiently large number to enforce

$$y_3 = 0 \Rightarrow t_3 = 0 \quad \text{and} \quad t_3 \geq 20000 \Rightarrow y_3 = 1$$

- This is a non-standard expression

Piecewise linear functions (Alternative)



■ Decision variables

$$y_i = \begin{cases} 1 & \text{if segment } i \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

- Segment 1: $z_1, z_2 \geq 0, z_1 + z_2 = 1, b = z_1 b_1 + z_2 b_2, f = z_1 f(b_1) + z_2 f(b_2)$
- Segment 2: $z_2, z_3 \geq 0, z_2 + z_3 = 1, b = z_2 b_2 + z_3 b_3, f = z_2 f(b_2) + z_3 f(b_3)$
- Segment 3: $z_3, z_4 \geq 0, z_3 + z_4 = 1, b = z_3 b_3 + z_4 b_4, f = z_3 f(b_3) + z_4 f(b_4)$

Piecewise linear functions (Alternative)

■ Decision variables

$$y_i = \begin{cases} 1 & \text{if segment } i \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

■ Segments:

■ Segment 1: $z_1, z_2 \geq 0, z_1 + z_2 = 1, b = z_1b_1 + z_2b_2, f = z_1f(b_1) + z_2f(b_2)$

■ Segment 2: $z_2, z_3 \geq 0, z_2 + z_3 = 1, b = z_2b_2 + z_3b_3, f = z_2f(b_2) + z_3f(b_3)$

■ Segment 3: $z_3, z_4 \geq 0, z_3 + z_4 = 1, b = z_3b_3 + z_4b_4, f = z_3f(b_3) + z_4f(b_4)$

■ Formulation:

$$z_1 \leq y_1, z_2 \leq y_1 + y_2, z_3 \leq y_2 + y_3, z_4 \leq y_3$$

$$y_1 + y_2 + y_3 = 1$$

$$z_1 + z_2 + z_3 + z_4 = 1$$

$$b = z_1b_1 + z_2b_2 + z_3b_3 + z_4b_4$$

$$f = z_1f(b_1) + z_2f(b_2) + z_3f(b_3) + z_4f(b_4)$$

$$z_1, z_2, z_3, z_4 \geq 0, y_1, y_2, y_3 \in \{0, 1\}$$

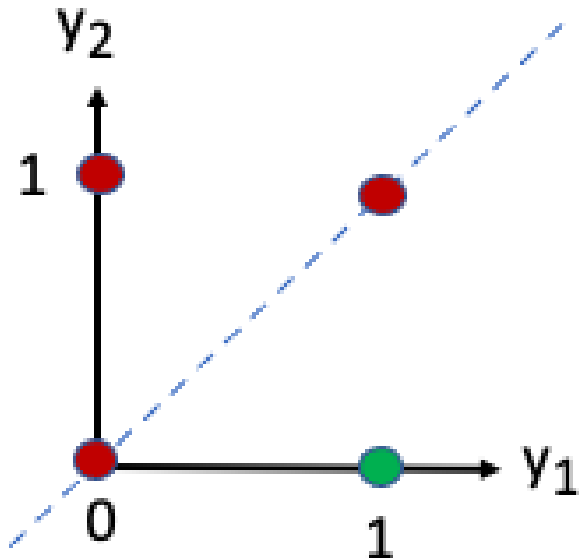
■ Reference: Winston, Chapter 9

Linear objective and constraints(1)

- All integer programming solvers assume the objective and constraints are linear in the decision variables
- A nonlinear objective can be approximated by piecewise linear functions

Linear objective and constraints(2)

- Nonlinear constraints can be replaced by linear constraints.
Example: Given that $y_1, y_2 \in \{0, 1\}$, we have
 - $y_1(1 - y_2) \geq 1$ can be replaced by $y_1 - y_2 > 0$ or $y_1 - y_2 \geq 1$



- Key point: Make the objective and constraints linear

This lecture: Summary

- Using binary integer variables to
 - Select an option out of many
 - Either-or requirement
 - If-then condition
 - Piecewise linear function
- Linear objectives and constraints

Integer programming and optimisation: Summary

- What you have learnt
 - How to formulate integer programming problems
 - How to solve them using AMPL
 - Examples of using integer programming for network design and analysis, placement problems
- There are a lot more to learn but this will give you a starting point
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References

- Advanced formulation of integer programming problems
 - Winston, “Operations Research”, Section 9.2